

Tzu-Ming Harry Hsu

徐子旻

AI researcher-engineer for embodied AI, ML systems, and product delivery

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Embodied AI, ML systems, and AI product delivery

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- MIT-trained researcher-engineer with 20+ publications, patents, and theses across computer vision, federated learning, sensing, medical AI, and production ML systems.
- Built large-scale experimentation infrastructure at Google spanning 1,000+ GPUs and 1M+ machine hours, with open-source code and dataset releases.
- Recent production work includes 10 TB data operations, labeling loops, deployment automation, throughput gains, and cost-aware ML platform improvements.
- Founder-operator experience raising \$1MM+, hiring and leading teams up to 20 people, and translating technical roadmaps into product delivery.

Education

Massachusetts Institute of Technology

Jun 2020 – Jun 2022

Ph.D., Computer Science

Cambridge, MA

- *Advisor:* Peter Szolovits
- *Research areas:* Computer vision, federated learning, machine learning for healthcare

Massachusetts Institute of Technology

Sep 2017 – May 2020

S.M., Electrical Engineering and Computer Science

Cambridge, MA

- *Advisors:* Peter Szolovits, Fadel Adib
- *Research areas:* Computer vision, machine learning, wireless signal, signal processing

National Taiwan University

Sep 2011 – Jun 2016

B.S., Physics; B.S.E., Electrical Engineering

Taipei, Taiwan

- *Class rank:* 1/190
- *GPA:* 3.99 / 4.00

Professional Experience

Clarq AI

Dec 2024 – Present

Co-founder & CTO

Palo Alto, CA / Taipei, Taiwan

Helped reposition Clarq from a hardware product into a force-intelligence platform, framing a long-term moat around force-time data, evaluation loops, model-driven product decisions, and fundraising.

- Co-led technical positioning around a three-layer strategy: hardware wedge, SaaS profit engine, and long-term data/model moat.
- Drove US-facing investor outreach and progressed pre-seed conversations toward near-commit stage with angels.
- Supported Startup Island Taiwan Silicon Valley program participation and international expansion narrative.
- Aligned product roadmap, data strategy, and investor-facing story across Taiwan hardware operations and US capital market expectations.

International Academia of Biomedical Innovation Technology

2021 – Present

Consultant & Convener

Taipei, Taiwan

Supported a 50+ member biomedical innovation community through project mentorship, dental AI research collaboration, and large-scale technology outreach events.

- Helped grow a community of 50+ members while supporting 8+ academic projects and 15+ publications.

- Led a dental AI research team in collaboration with National Taiwan University Hospital.
- Co-organized demo events with 100+ participants to showcase biomedical technologies and educate the broader community.

Dentscape

Machine Learning & Software Engineering Consultant

Apr 2025 – Apr 2026

San Francisco, CA / Taipei, Taiwan

Built data, training, and evaluation systems for AI-assisted dental design, combining geometric ML, pre-labeling workflows, preference modeling, and production deployment improvements.

- Reorganized a fragmented ~10 TB dental dataset into a standardized ~4 TB ML-ready registry, reducing raw-data access from ~3 days to minutes.
- Rebuilt pre-labeling and labeling loops for 100-200 case batches, reducing turnaround from ~4 weeks to ~1 week and improving experiment velocity.
- Implemented bilevel/meta-learning for personalized crown generation and improved preference fit by ~20% while keeping the system production-facing.
- Optimized segmentation and crown-generation services for ~110% aggregate throughput improvement while reducing infrastructure cost by ~50%.

CodeGreen Labs PBC

Co-founder & CTO

Sep 2023 – Nov 2024

Boston, MA / Taipei, Taiwan

Led a 10-person engineering organization delivering auditable data systems for enterprise partners, reaching projected \$1MM annual revenue while hardening integration, compliance, and delivery operations.

- Reached projected annual revenue of \$1MM USD by Q2 2024 with a 10-person team.
- Established partnerships with Microsoft, Global Blockchain Business Council, Gold Standard, SDG Data Alliance, and PVBLIC.
- Built a cross-functional team across product, partnerships, and engineering to address complex data, auditability, and compliance challenges.
- Designed and shipped data-intensive systems with end-to-end auditability, regulatory compliance, and interoperability with existing partner ecosystems.

Hashgreen Labs

Co-founder & CTO

Sep 2022 – Aug 2023

Boston, MA / Taipei, Taiwan

Built and led a 20-person engineering organization, raised \$1MM+ in capital, and shipped client-facing data and transaction systems while establishing hiring, delivery, and product operating cadence from zero.

- Raised over \$1MM USD in capital and grew the team from 0 to 20 employees.
- Built hiring, onboarding, and performance review systems to support a high-performing engineering organization.
- Secured and delivered high-impact projects for clients including The World Bank and Chia Network.
- Led development of DeFi protocols (DEX and AMM), becoming the first team on the Chia blockchain to launch these primitives.

Massachusetts Institute of Technology

Research Assistant

Sep 2018 – Jun 2022

Cambridge, MA

Led research on deployable medical AI, computer vision, and learning under data constraints, producing 20+ publications, patents, and theses with translational partners across industry and healthcare.

- Published 14 conference articles, 5 journal articles, one U.S. patent, and two degree theses.
- Served as a peer reviewer for 8 conferences and journals, reviewing 34 articles.
- Received media coverage from CBS, Engadget, TechCrunch, and MIT News.
- Collaborated with Google, Takeda Pharmaceutical, and Beth Israel Deaconess Medical Center.
- Awarded scholarships totaling 4 years in duration.

WorldQuant

Data Science Intern

Jun 2021 – Aug 2021

Taipei, Taiwan

Improved an existing quantitative strategy by +1.5% market correlation through targeted modeling and optimization work.

Beth Israel Deaconess Medical Center

Nov 2019 – Oct 2020

*Research Trainee**Boston, MA*

Published impactful clinical AI research and deployed a PACS-integrated screening workflow across 1,000+ patients while managing shared deep-learning compute for a 10+ member research unit.

- Published three impactful research articles on critical medical topics, including liver tumor assessment, COVID-19 clinical risk prediction, and automatic pancreatic cancer assessment.
- Developed and implemented an AI-driven patient screening pipeline in the hospital's PACS workflow, screening over 1,000 patients and improving disease diagnosis accuracy.
- Managed the computation resources for a research unit of 10+ members, streamlining deep learning processes and maximizing resource sharing.

Google

Jun 2020 – Sep 2020

*Software Engineer Intern**Remote*

Contributed production software and experimentation tooling that improved reliability and iteration speed for applied machine learning workflows.

Google

Jun 2019 – Mar 2020

*Researcher**Seattle, WA / Cambridge, MA*

Built large-scale ML experimentation infrastructure spanning 1,000+ GPUs and 1M+ machine hours, and published 2 peer-reviewed papers in deep federated learning with open-source outputs for broader adoption.

- Conducted foundational research in deep federated learning, contributing to early growth of the field.
- Published 2 peer-reviewed papers and presented findings at top conferences and workshops.
- Developed a scalable experimentation library running on thousands of machines and 1,000+ GPUs (over 1M machine hours).
- Open-sourced research code and datasets for broader community use.

Digital Drift

Mar 2016 – Mar 2020

*Lead Machine Learning Scientist**Taipei, Taiwan*

Built and deployed deep learning systems and backend APIs that moved computer vision features from experimentation into production mobile experiences.

- Built deep neural networks for cuisine recognition and matching using TensorFlow on multi-GPU machines.
- Designed and maintained backend APIs to serve model predictions in production settings.

Brigham and Women's Hospital

Mar 2019 – Feb 2020

*Research Trainee**Boston, MA*

Contributed hospital-facing medical-imaging ML work on risk prediction and deployment-oriented evaluation using real patient cohorts and radiology workflows.

- Supported model development and validation analysis for clinical imaging applications tied to operational hospital use cases.

Academia Sinica

Feb 2014 – May 2016

*Student Researcher**Taipei City, Taiwan*

Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification and published three peer-reviewed papers presented at top conferences and workshops.

- Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification.
- Published 3 peer-reviewed academic papers and presented findings at top conferences and workshops.

Olympiad Tutoring Community

Sep 2011 – Jun 2015

*Private Tutor**Taipei, Taiwan*

Mentored advanced physics students from fundamentals through competition-level performance, including pathways to Taiwan's national Olympiad representation.

- Mentored students who became Taiwan national representatives in the International Physics Olympiad (IPhO).

Leadership Experiences

Federation of Taiwanese Student Associations in New England (FT-SANE) <i>Activities Officer</i>	2019 – 2020 <i>Boston, MA</i>
MIT Taiwanese Student Association <i>President</i>	2018 – 2019 <i>Boston, MA</i>
• Led coordination of 20+ events for a diverse community of over 100 members. • Organized 10+ career workshops and helped plan 10+ national holiday celebrations.	

Awards & Honors

Departmental Valedictorian (First Place at Graduation) <i>Department of Electrical Engineering, National Taiwan University</i>	2016 <i>Taipei, Taiwan</i>
• Placed 1st among 200 classmates across all courses.	
Silver Medal <i>Altera Innovate Asia FPGA Design Competition</i>	2015 <i>Taipei, Taiwan</i>
• Placed 2nd among 20 international teams. • Designed a product named EZBud with integrated algorithms that modulate music based on user sporting statistics.	
Gold Medal / Overall Winner <i>42nd International Physics Olympiad (IPhO)</i>	2011 <i>Bangkok, Thailand</i>
• Represented Taiwan and placed 1st in theory, experiment, and combined rankings among 400+ participants from 80+ countries.	
Gold Medal / Experiment Winner <i>12th Asian Physics Olympiad (APhO)</i>	2011 <i>Tel Aviv, Israel</i>
• Placed 1st in the experiment section among 100+ participants.	
Gold Medal <i>41st International Physics Olympiad (IPhO)</i>	2010 <i>Zagreb, Croatia</i>
Gold Medal / Experiment Winner <i>11th Asian Physics Olympiad (APhO)</i>	2010 <i>Taipei, Taiwan</i>
• Placed 1st in the experiment section among 100+ participants.	
Gold Medal <i>5th International Junior Science Olympiad (IJSO)</i>	2008 <i>Changwon, Korea</i>

- Performance Frontier in Freediving: An Exploratory Analysis** 2026
AIDA Instructor Thesis
 TMH Hsu, J Chang
- Artificial Intelligence to Assess Dental Findings from Panoramic Radiographs—A Multinational Study** 2025
arXiv Cited by 8
 Yin-Chih Chelsea Wang, Tsao-Lun Chen, Shankeeth Vinayahalingam, Tai-Hsien Wu, Chu Wei Chang, Hsuan Hao Chang, Hung-Jen Wei, Mu-Hsiung Chen, Ching-Chang Ko, David Anssari Moin, Bram van Ginneken, Tong Xi, Hsiao-Cheng Tsai, Min-Huey Chen, Tzu-Ming Harry Hsu, Hye Chou
- Intra-Oral Scan Segmentation Using Deep Learning** 2023
BMC Oral Health Cited by 42
 Shankeeth Vinayahalingam, Steven Kempers, Julian Schoep, Tzu-Ming Harry Hsu, David Anssari Moin, Bram van Ginneken, Tabea Flugge, Marcel Hanisch, Tong Xi
- Positional Assessment of Lower Third Molar and Mandibular Canal Using Explainable Artificial Intelligence** 2023
Journal of Dentistry Cited by 29
 Steven Kempers, Pieter van Lierop, Tzu-Ming Harry Hsu, David Anssari Moin, Stefaan Berge, Hossein Ghaemina, Tong Xi, Shankeeth Vinayahalingam
- Emulating Clinical Diagnostic Reasoning for Jaw Cysts with Machine Learning** 2022
Diagnostics Cited by 22
 Balazs Feher, Ulrike Kuchler, Falk Schwendicke, Lisa Schneider, Jose Eduardo Cejudo Grano de Oro, Tong Xi, Shankeeth Vinayahalingam, Tzu-Ming Harry Hsu, Janet Brinz, Akhilanand Chaurasia
- Methods and Apparatus for Radio Frequency Sensing in Diverse Environments** 2022
US Patent 11,308,291 Cited by 16
 Unsoo Ha, Junshan Leng, Alaa Khaddaj, Yunfei Ma, Tzu Ming Hsu, Zexuan Zhong, Fadel Adib
- Effective Modeling in Medical Imaging with Constrained Data** 2022
PhD Thesis, Massachusetts Institute of Technology
 Tzu-Ming Harry Hsu
- Artificial Intelligence to Assess Body Composition on Routine Abdominal CT Scans and Predict Mortality in Pancreatic Cancer – A Recipe for Your Local Application** 2021
European Journal of Radiology Cited by 57
 Tzu-Ming Harry Hsu, Khoschy Schawkat, Seth J Berkowitz, Jesse L Wei, Alina Makoyeva, Kaila Legare, Corinne DeCicco, S Nicolas Paez, Jim SH Wu, Peter Szolovits
- Visceral Adiposity and Severe COVID-19 Disease: Application of an Artificial Intelligence Algorithm to Improve Clinical Risk Prediction** 2021
Open Forum Infectious Diseases Cited by 23
 Alexander Goehler, Tzu-Ming Harry Hsu, Jacqueline A Seigle, Mark J Siedner, Janet Lo, Virginia Triant, John Hsu, Andrea Foulkes, Ingrid Bassett, Ramin Khorasani, Deborah J Wexler, Peter Szolovits, James B Meigs, Jennifer Manne-Goehler
- Adversarial Contrastive Pre-Training for Protein Sequences** 2021
arXiv Preprint arXiv:2102.00466 Cited by 9
 Matthew McDermott, Brendan Yap, Harry Hsu, Di Jin, Peter Szolovits

DeepOPG: Improving Orthopantomogram Finding Summarization with Weak Supervision	2021
<i>Medical Image Computing and Computer Assisted Intervention (MICCAI 2021)</i>	<i>Cited by 8</i>
Tzu-Ming Harry Hsu, Yin-Chih Chelsea Wang	
Methods and Apparatus for Radio Frequency Sensing in Diverse Environments	2020
<i>US Patent 10,872,209</i>	<i>Cited by 16</i>
Unsoo Ha, Junshan Leng, Alaa Khaddaj, Yunfei Ma, Tzu Ming Hsu, Zexuan Zhong, Fadel Adib	
Three-Dimensional Neural Network to Automatically Assess Liver Tumor Burden Change on Consecutive Liver MRIs	2020
<i>Journal of the American College of Radiology</i>	<i>Cited by 23</i>
Alexander Goehler, Tzu-Ming Harry Hsu, Ronilda Lacson, Isha Gujrathi, Raein Hashemi, Grzegorz Chlebus, Peter Szolovits, Ramin Khorasani	
Chexpert++: Approximating the Chexpert Labeler for Speed, Differentiability, and Probabilistic Output	2020
<i>Machine Learning for Healthcare Conference (MLHC 2020)</i>	<i>Cited by 45</i>
Matthew BA McDermott, Tzu Ming Harry Hsu, Wei-Hung Weng, Marzyeh Ghassemi, Peter Szolovits	
Automatic Longitudinal Assessment of Tumor Responses	2020
<i>PhD Thesis, Massachusetts Institute of Technology</i>	<i>Cited by 3</i>
Tzu-Ming Harry Hsu	
Baselines for Chest X-Ray Report Generation	2020
<i>Machine Learning for Health (ML4H) Workshop, NeurIPS 2020</i>	<i>Cited by 94</i>
William Boag, Tzu-Ming Harry Hsu, Matthew McDermott, Gabriela Berner, Emily Alesentzer, Peter Szolovits	
Federated Visual Classification with Real-World Data Distribution	2020
<i>European Conference on Computer Vision (ECCV 2020)</i>	<i>Cited by 275</i>
Tzu-Ming Harry Hsu, Hang Qi, Matthew Brown	
Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification	2019
<i>arXiv Preprint arXiv:1909.06335</i>	<i>Cited by 1972</i>
Tzu-Ming Harry Hsu, Hang Qi, Matthew Brown	
Transfer Neural Trees: Semi-Supervised Heterogeneous Domain Adaptation and Beyond	2019
<i>IEEE Transactions on Image Processing</i>	<i>Cited by 21</i>
Wei-Yu Chen, Tzu-Ming Harry Hsu, Yao-Hung Hubert Tsai, Ming-Syan Chen, Yu-Chiang Frank Wang	
Clinically Accurate Chest X-Ray Report Generation	2019
<i>Machine Learning for Healthcare Conference (MLHC 2019)</i>	<i>Cited by 398</i>
Guanxiong Liu, Tzu-Ming Harry Hsu, Matthew McDermott, Willie Boag, Wei-Hung Weng, Peter Szolovits, Marzyeh Ghassemi	
Unsupervised Multimodal Representation Learning Across Medical Images and Reports	2018
<i>Machine Learning for Health (ML4H) Workshop, NeurIPS 2018</i>	<i>Cited by 51</i>
Tzu-Ming Harry Hsu, Wei-Hung Weng, Willie Boag, Matthew McDermott, Peter Szolovits	
Learning Food Quality and Safety from Wireless Stickers	2018
<i>Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets 2018)</i>	<i>Cited by 95</i>
Unsoo Ha, Yunfei Ma, Zexuan Zhong, Tzu-Ming Hsu, Fadel Adib	

- 3D-Aware Scene Manipulation via Inverse Graphics** 2018
Advances in Neural Information Processing Systems (NeurIPS 2018) Cited by 321
 Shunyu Yao, **Tzu Ming Hsu**, Jun-Yan Zhu, Jiajun Wu, Antonio Torralba, Bill Freeman, Josh Tenenbaum
- Transfer Neural Trees for Heterogeneous Domain Adaptation** 2016
European Conference on Computer Vision (ECCV 2016) Cited by 80
 Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Yao-Hung Hubert Tsai, Yu-Chiang Frank Wang, Ming-Syan Chen
- Connecting the Dots Without Clues: Unsupervised Domain Adaptation for Cross-Domain Visual Classification** 2015
International Conference on Image Processing (ICIP 2015) Cited by 2
 Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Cheng-An Hou, Yi-Ren Yeh, Yu-Chiang Frank Wang
- Unsupervised Domain Adaptation with Imbalanced Cross-Domain Data** 2015
International Conference on Computer Vision (ICCV 2015) Cited by 90
Tzu Ming Harry Hsu, Wei Yu Chen, Cheng-An Hou, Yao-Hung Hubert Tsai, Yi-Ren Yeh, Yu-Chiang Frank Wang